HIMANSHU SHUKLA

MACHINE LEARNING ENGINEERING / APPLIED AI

Gwalior, Madhya Pradesh | himanshusubh1510@gmail.com | GitHub | LinkedIn | Portfolio

PROFESSIONAL PROFILE

Mathematics & Computing undergraduate exploring cross-modal image retrieval, embedding search, evaluation methodology and computer-vision prototypes. Interested in measurable experiments and reproducible validation.

EDUCATION

Madhav Institute of Technology & Science (MITS), Gwalior | B.Tech Mathematics & Computing | Expected 2028 | CGPA 7.75/10 (through Semester IV)

TECHNICAL SKILLS

Machine learning: PyTorch, scikit-learn, NumPy, Pandas, contrastive learning, embeddings, FAISS, F1@K
Programming: Python, C++, MATLAB, SQL, PostgreSQL
Experiments & tools: FastAPI, pytest, Docker, Jupyter, Git/GitHub, manifest validation, latency measurement

SELECTED PROJECTS

EarthBridge | **Cross-modal satellite-image retrieval | PyTorch, FAISS**

- Developed a research prototype for optical, SAR and multispectral descriptor retrieval with contrastive representations and a FastAPI interface.
- Implemented F1@5/F1@10 evaluation procedures, leakage-aware splits, latency checks and a 128-pair overfit gate; no unverified performance metric claimed.

IrisTrack | **Experimental opt-in webcam tracking | Computer vision**

- Developed face/iris landmarks, blink and head-pose quality checks, calibration and validation flows; real-webcam accuracy and installer readiness remain unverified.

Mulyankan | **Evidence-linked evaluation | React, TypeScript, Supabase**

- Designed evidence references, insufficient-evidence flags and human reviewer controls; demonstrated the prototype at CMPDI in July 2026.

Sentinel | **Local flow monitoring | Python, Scapy, FastAPI**

- Created packet/flow monitoring views and an early anomaly-analysis interface without claiming production intrusion-detection performance.

ADDITIONAL WORK

NewsSources (article/PDF ingestion and provenance); OHS (React/Firebase collaboration); Mulyankan Spatial Lab (3D explanatory interface).

LEADERSHIP & RECOGNITION

Class Representative (3rd Year); NPTEL Coordinator; Training & Placement Cell member; SchoolBuddies co-founder. First place - Neeti Summit English Debate.